

Renewable energy communities for sustainable cities: Economic insights into subsidies, market dynamics and benefits distribution

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HIGHLIGHTS

- PV plants within RECs have high returns and low levels of risk
- Two innovative Profit Distribution Models are proposed in this study
- These models promote energy-efficient behaviors and address income inequality
- Sustainable cities can be enhanced by the presence of renewable energy plants
- The development of RECs is a vital tool for SDGs 7 and 11

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ABSTRACT

Renewable energy communities (RECs) present a compelling pathway to achieving a just energy transition, particularly in regions such as Europe. Within this context, Italy stands out as a leader in REC development due to its favorable incentive policies, which support widespread adoption. The advancement of RECs requires robust quantitative analyses to illustrate the equitable distribution of economic benefits between prosumers and consumers. This paper presents an economic evaluation of photovoltaic (PV) systems within residential RECs, contributing to the development of sustainable and green urban environments. Using the discounted cash flow method with net present value (NPV) as the primary indicator, the research evaluates various scenarios influenced by subsidies and market dynamics. The findings reveal that PV-based RECs represent highly profitable investments characterized by low economic risk. The NPVs per kW range from €2042–5148 in low-market scenarios and €3347–8195 in high-market scenarios. For RECs comprising four prosumers, the distribution of benefits varies between 15 and 35 %. Additionally, the study introduces two innovative profit-sharing models among renewable self-consumers (RSCs). The first model incentivizes energy-conscious consumption by offering lower trading prices within the REC, while the second model prioritizes energy equity by providing greater support to lower-income households. Both models result in an 18–30 % reduction that encourages prosumer participation, irrespective of individual consumption patterns. Furthermore, the inclusion of consumers alters the distribution of benefits, reinforcing the role played by RECs in fostering socially inclusive energy systems. Ultimately, the economic viability of PV systems within RECs underscores their critical role in advancing energy-sustainable urban development. By simultaneously enhancing environmental resilience and promoting social equity, RECs may serve as a foundational element in the transition towards a greener and more equitable energy future.

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Nomenclature			
Acronym		r_d	Loan interest rate
GME	“Gestore Mercati Energetici”	S	Plant size
GSE	“Gestore Servizi Energetici”	S_{REC}^u	Unitary subsidy REC
HM	High market	$Sp_{el,t}$	Sale of energy
LCOE	Levelized cost of electricity	t	Time
LM	Low market	t_r	Average annual insolation
MASE	“Ministero dell'Ambiente e della Sicurezza Energetica”	$TaxD_u$	Unitary tax deduction
NPV	Net present value	Vat	Value added tax
PDM	Profit distribution model	$\omega_{self,c}$	Percentage of self-consumed energy
PV	Photovoltaic	T_{GSE}	GSE fees
REC	Renewable energy community	$T_{GSE,rid}$	Fee due to GSE dedicated withdrawal
RO	Research objective	Symbol	
RSC	Renewable self-consumer	E_{out}	Energy production
SDG	Sustainable development goal	E_p	Energy produced by each RSC
Symbol		E_{PNSCj}	Partial energy not self-consumed by each RSC
A_{cell}	Active surface	E_{PSCj}	Partial energy self-consumed by each RSC
C_{ae}	Administrative/electrical connection cost	E_r	Energy required by the additional consumer to the REC
C_{inv}	Investment cost	j	Single RSC
$C_{inv,unit}$	Unitary investment cost	N_{RSC}	Number of RSCs
C_{lcs}	Loan capital share cost	NPV_{RSCj}	NPV associated with the j-th RSC
C_{val}	Unit enhancement fee	OB_j	Overall benefits of each RSC
DCI	Discounted cash inflows	OB_{TOT}	Overall benefits
DCO	Discounted cash outflows	p^c	Electricity purchase price
dE_f	Decreased system efficiency	p^{ex}	Exchange price
E_{out}	Energy production	p^s	Electricity selling price
inf	Rate of inflation	p_p	Price proposed by each RSC
inf_{el}	Rate of energy inflation	p_v	Associated weight according to virtuous behavior (or minimum incomes)
k_f	Optimum tilt angle	PD_{OV}	Percentage of the benefits of each RSC
N	PV system lifetime	S_{EX}	Economic savings associated with exchange between RSCs
N_{debt}	Loan period	S_{EXv}	Economic savings associated with exchange between RSCs as a function of virtuous behavior (or minimum incomes)
N_{TaxD}	Deduction period	S_{Net}	Sale of non-produced and non-self-consuming energy to the grid by each RSC
η_{bos}	Balance of system efficiency	S_s	Incentives from self-consumption of energy
η_m	Module efficiency	S_{SC}	Savings from self-consumed energy by each RSC
p^c	Electricity purchase price	S_u	Energy community incentive + enhancement fee
P_{Cass}	Percentage of assurance cost	$\omega_{self,cj}$	Value of self-consumption associated with the j-th RSC
P_{Ci}	Percentage of inverter cost	$\varpi_{self,c}$	Average value of self-consumption
P_{Cm}	Percentage of maintenance cost	ω_{vj}	Percentage benefits based on virtuous behavior (or minimum incomes)
P_{Ctax}	Percentage of taxes cost		
p^{ex}	Exchange price		
p^s	Electricity selling price		
r	Opportunity cost of capital		

1. Introduction

Urban development must align local renewable energy strategies with national and global energy systems [1]. To this end, sustainability assessments of cities and communities are essential [2]. Policymakers and stakeholders must prioritize the Sustainable Development Goals (SDGs), and particularly SDG 11, to design cities that address the needs of local communities, attract tourists and contribute to global climate stability. In this process, the proper sizing of facilities in residential communities is crucial to deliver significant energy, financial and environmental benefits [3]. In addition, SDG 7 emphasizes the transformation of energy systems towards sustainable development [4], creating synergies with other SDGs [5,6].

Smart cities, defined by economic activities fostering sustainable growth [7], impose new challenges on urban planners, such as the integration of artificial intelligence to enhance community efficiency [8,9]. The urban energy transition requires a balance to be struck between buildings and open spaces, as well as the optimization of

renewable system installations [10]. Prosumers, encompassing various participant types [11], are foundational to sustainable urban transformation [12]. Therefore, effective urban planning must be closely linked to energy considerations [13].

Energy communities (ECs) are defined as collections of actors who self-produce energy using systems in close proximity, though not necessarily within the same building [14]. European policies recognize renewable energy communities (RECs) as pivotal to achieving a low-carbon society [15], supporting SDGs 7, 11 and 13 [16]. RECs offer environmental and economic benefits [17], driven by citizen participation. While environmental and climate motivations play a role in determining this participation [18], economic savings on energy costs are typically the primary driver. Moreover, additional factors, such as the removal of technical barriers and enhanced energy independence, influence green energy transition choices [19].

Policy frameworks are crucial for ensuring a fair and equitable energy transition. An absence of subsidies remains a significant obstacle to the adoption of PV systems [20], and a lack of public support further

deters their installation [21]. Legal complexities can also present barriers for prosumers [22], highlighting the need for clear regulation [23]. Socio-institutional factors are equally critical in fostering ECs [24].

Social opposition phenomena such as “Not in My Back Yard” and “Not in My Term of Office” can significantly delay project implementation and jeopardize energy independence—a critical aspect of national security and competitiveness [25]. Furthermore, social trust cannot be underestimated, as some analysts have observed that structural interactions with neighbors and civic norms within families positively influence willingness to participate in an REC [26]. Therefore, democratic participation and energy innovation are highly desirable for REC development [27]. An additional goal of RECs is to combat energy poverty through enhanced accessibility [28,29], with multiple studies demonstrating the tangible benefits of these efforts [30,31].

PV-based RECs represent a new social model for the energy transition, aligned with the principles of the green economy [32]. However, challenges arise when some REC members prioritize profit over community goals [33], necessitating a careful balance to be struck between individual households' financial expectations and the community's self-sufficiency objectives [34]. However when environmental priorities are emphasized, there is greater potential to optimize PV system use [35], further supported by the integration of energy storage systems [36,37] and programmable electrical appliances [38]. Nevertheless, compromise-seeking processes, if poorly managed, can undermine community stability [39]. Models must ensure equitable treatment of smaller community members [40] and balance the interactions between ECs and distribution grids [41].

Two central aspects of energy sharing emerge from the literature: internal regulations governing the distribution of benefits [42] and external regulations establishing the overarching regulatory framework [43]. The equitable distribution of benefits among renewable self-consumers (RSCs) is a recurring theme in the literature, and thus a robust sharing of risks and benefits is critical [44]. Various approaches have been proposed to define benefits-sharing mechanisms among RSCs [32], including models incorporating consumers [25] or new participants in multiple RECs [45], and mathematical techniques and neural networks to optimize outcomes [46]. Net present value (NPV) is frequently used as the objective function in these analyses [45], with economic savings prioritized as a key factor driving REC participation [19].

A recent review of RECs identified four primary areas of scholarly focus: (i) economic and financial considerations within regulatory frameworks, (ii) RECs as organizational structures, (iii) PV systems as predominant energy sources and (iv) Italy as a leading case study. The review also underscored the critical need to evaluate benefits-sharing mechanisms among community members [47]. The present study sought to address this research gap by proposing practical business models for real-world application. The analysis included an evaluation of the profitability of a residential PV system in Italy, followed by an exploration of innovative benefits-sharing models for RSCs. Baseline and alternative scenarios were examined, including the integration of a consumer (non-producer) into the community framework.

The study introduced two novel models for benefits distribution within RSCs. The first model-based allocation on the trading price proposed within the community, encouraging equitable participation. The second model incorporated income brackets, explicitly targeting energy poverty and fostering inclusivity. These models aimed at providing actionable frameworks for balancing economic and social priorities in RECs.

The remainder of the paper is structured as follows. Section 2 presents a review of the literature, incorporating the PRISMA model to synthesize research on REC-related economic issues. Section 3 describes the profitability assessment of a residential PV system in Italy, introducing a new model for allocating premiums to RSCs, based on either virtuous behavior (e.g., proposing the lowest exchange price within the community) or income-based criteria to support lower-income

participants. Section 4 describes the methodology, which employed NPV as the primary indicator of profitability under both baseline and alternative scenarios. Key variables included the purchase price of energy, the energy selling price and the plant investment cost (Section 4). Section 5 compares the proposed benefits-sharing models with existing models in the literature. Beyond the baseline case, alternative scenarios are analyzed, including variations in individual RSC self-consumption rates and the introduction of a consumer (non-producer) into the REC. Section 6 concludes the paper, summarizing the key findings and implications.

2. Literature review

Amid the growing focus on sustainability and energy efficiency, the roles played by RECs and collective participation in energy production have gained increased attention. The present systematic literature review (SLR) focused on economic analyses of these phenomena using metrics such as NPV, profitability and economic savings, adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [48].

The Scopus database was employed for the article search on September 6, 2024. Search terms included “energy communities” and “collective self-consumption,” combined with economic keywords such as “NPV,” “economic analysis,” “profitability analysis” and “economic saving.” The initial query yielded 128 articles published between 2020 and 2024. The specific search query was as follows:

TITLE-ABS-KEY ((“Energy communities” OR “Collective self-consumption”) AND (“NPV” OR “Economic analysis” OR “Profitability analysis” OR “Economic saving”)) AND PUBYEAR > 2019 AND PUBYEAR < 2025.

After applying filters to include only articles and reviews written in English and published in academic journals, the selection was reduced to 98 articles. Two exclusion criteria were applied:

- E1: The publication was not an article or review.
- E2: The publication was not in English.

Publication trends, as presented in Table S1, showed a steady increase in research activity, peaking in 2023 (33 articles). Despite only partial data for 2024 (20 articles already published), the trend indicated sustained interest in RECs. Key contributors to the field included Francesca Ceglia and Elisa Marrasso (University of Sannio), as well as Idiano D'Adamo (Sapienza University of Rome), who collectively co-authored several papers. The selection of 98 articles spanned a diverse range of journals, with *Energies* leading the field (19 articles), followed by *Renewable Energy* (8 articles), *Sustainability*, *Energy Conversion and Management* and *Applied Energy* (Fig. S1).

Geographically, Italy dominated the research landscape (42 articles), likely reflecting its strong institutional and regulatory support for ECs. Other notable contributors included Spain and China, highlighting a global interest in the topic with a pronounced European focus.

Using the Bibliometrix package in RStudio, a word cloud (Fig. S2) was generated to highlight the main key terms in the reviewed studies, which included “energy,” “community,” “communities” and “renewable.” The prominence of “renewable,” in particular, highlights the critical connection between ECs and renewable energy sources, which are essential for sustainability and collective self-consumption. Moreover, economic terms such as “economic,” “profitability” and “saving” indicated a significant focus on financial benefits and profitability models. Technological themes included “photovoltaic,” “storage,” “self-consumption” and “hydrogen,” reflecting growing interest in technological solutions for integrating renewables and supporting self-consumption. Finally, consistent with Italy's dominance in publication volume, the term “Italy” appeared frequently in the articles.

The thematic map presented in Fig. 1 further explores the key themes of the identified articles, offering an analytical perspective on the interconnected topics within the literature.

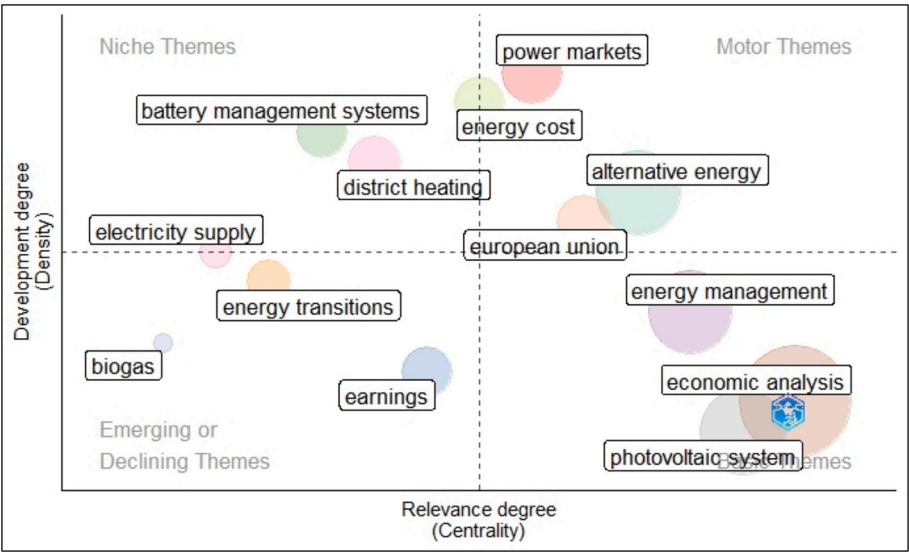


Fig. 1. Thematic map.

The axes are defined according to thematic relevance (centrality) and development (density):

- Motor themes are central and well developed. Among these, the European Union stands prominently, indicating its key role in promoting and developing a decarbonized energy model. Of

note, the focus on alternative energies underscores Europe's active efforts to integrate alternative energies (particularly renewables) into its development plan, which also encompasses ECs. In this direction, the energy buying and selling supply chain is evolving to include a broader range of participants (particularly prosumers, who may differ from the large national

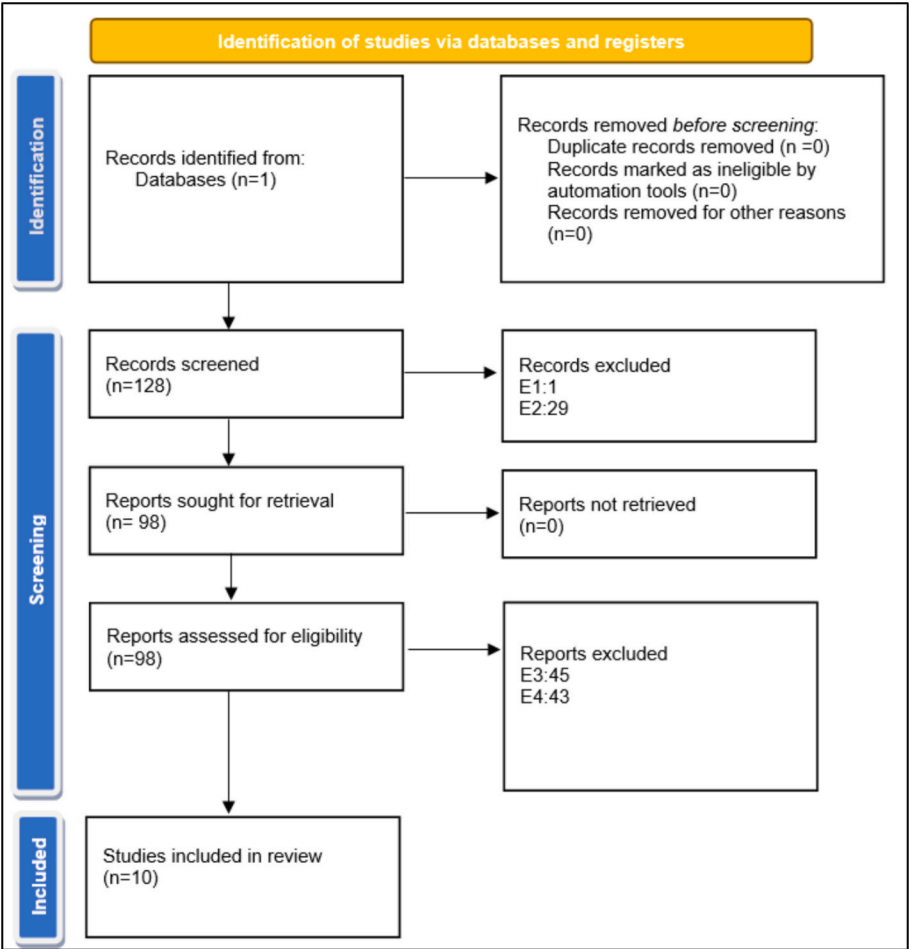


Fig. 2. Document selection process.

energy players) and innovative models (as highlighted by the theme of power markets).

- Basic themes include economic analysis and PV systems. These themes are highly relevant but exhibit lower levels of recent development, suggesting that they are established areas of discourse with less ongoing innovation.
- Niche themes include battery management systems (crucial for energy stability) and district heating (important for urban decarbonization). These themes reflect significant technological development with strong academic and industrial interest. However, they remain peripheral to broader economic discussions regarding collective energy consumption.
- Emerging and declining themes focus on the energy transition, and particularly the shift from fossil fuels to green energy sources.

Given the observed emphasis on solar and PV technologies, the scope was refined to focus specifically on these technologies. This narrowed the selection to 53 articles, revealing that not all of the investigated ECs relied exclusively on these technologies.

- E3: The publication did not address solar and PV technologies.

Using the “Relevance” function on Scopus, the remaining articles were sorted by relevance and the top 10 were selected for further analysis (Fig. 2).

- E4: The publication was not among the 10 most relevant.

Among the 10 papers analyzed, various approaches were used to address different aspects of RECs and collective energy consumption. One of the first studies focused on the integration of RECs in public residential buildings in northern Spain, demonstrating that the installation of PV systems may result in a very short economic payback period of approximately 2 years and significant CO₂ savings, estimated at about 1600 tons annually [49].

In Italy, an analysis of the profitability of a PV system under a collective self-consumption scenario revealed that incentive policies could significantly impact NPV, with variations ranging from 2136 to 8084 €/kW in the baseline scenario and from 1919 to 7868 €/kW in an alternative policy scenario. The introduction of an innovative benefits-sharing model underscored how such an approach may render ECs sustainable from both economic and social perspectives [32]. Similarly, other studies emphasized the importance of equitable benefits-sharing mechanisms and the promotion of virtuous citizen behaviors to foster sustainable cities powered by green energy [25]. A further study of the Italian territory explored the optimization of self-consumption and energy self-sufficiency using business models combining PV systems with electrochemical storage. This research indicated that a community in northwestern Italy could achieve an internal efficiency improvement of more than 11 % while reducing CO₂ emissions by approximately 45 % [50].

With respect to advanced technology, the impact of smart metering and smart charging within an REC equipped with electric vehicles was analyzed, showing that these technologies may significantly enhance self-consumption and reduce energy costs [51]. Another study extended beyond PV systems, also exploring the integration of renewable energy sources such as biomass cogeneration and mini-hydro power. The results demonstrated increased overall energy production, with an economic payback period of approximately 5 years and an NPV ranging from 12.4 to 13.3 million € [52]. A further comparative analysis of post-pandemic policies in Italy and Portugal revealed differing outcomes: Italian policies made PV systems with storage particularly profitable, while Portuguese policies favored smaller PV systems without storage. In both contexts, RECs consistently emerged as more economically advantageous than individual self-consumption models [53].

Another interesting aspect explored in the literature was the impact of second-life batteries for RECs in Italy. Research showed that these batteries, repurposed from electric vehicles, may enhance solar energy self-consumption and optimize demand-side management while reducing utility bills and promoting a circular approach to energy use [54]. Additionally, a study analyzing the performance of different PV systems within a community of buildings yielded promising results in terms of energy self-sufficiency and self-consumption. With a net energy balance of 139 % and a payback period ranging from 9 to 17 years, the findings underscored the economic sustainability of energy transition efforts within well-designed communities [55].

Finally, research into electric pricing and local energy system optimization demonstrated significant increases in installed PV capacity, accompanied by notable economic benefits. However, it also highlighted the risks associated with the uncoordinated design of RECs, which can lead to system cost increases of up to 83 % due to inefficient use of local renewable energy sources [56].

In conclusion, the reviewed studies reinforced the potential of RECs as a model for the green transition, offering substantial benefits in terms of both environmental sustainability and economic viability. However, the effective implementation of RECs necessitates a thorough evaluation of benefits-sharing mechanisms, which is missing in the literature. This a critical gap that the present study aimed to address [44,47]. The existing literature predominantly presents models that distribute benefits either equally—irrespective of individual energy consumption profiles—or strictly based on consumer profiles. However, these approaches (described as extreme cases) may undermine the practical implementation of RECs by failing to account for the complexities of equitable participation.

Previous studies have also explored scenarios in which the energy exchange price is calculated as the average between the energy purchase and selling prices [25,32]. Building on these insights, the present work advocates for the development of innovative business models that balance economic feasibility with inclusivity, addressing the limitations of existing frameworks and supporting the sustainable implementation of RECs.

3. Research design

Following a brief overview of the governance structure of the Italian energy system and the regulations concerning RECs (Section 3.1), Section 3.2 proposes the NPV-based economic model. This is followed by an analysis of several case studies (Section 3.3). Section 3.4 details the input data used to evaluate the cost-effectiveness of the investment, while Section 3.5 aims to define the benefits-sharing models used by the various RECs. The analyses refer to a specific area in central Italy.

3.1. Italian regulatory system

An REC is a legal entity comprising members such as citizens, small and medium-sized enterprises (SMEs) (for whom participation in the REC is not a primary commercial or industrial activity) and local authorities. Within an REC, renewable electricity is shared among producers and consumers connected to the same primary substation, and the national electricity distribution grid is utilized to facilitate energy sharing. REC members can apply to Italy's Energy and Climate Agency, the Gestore dei Servizi Energetici (GSE), to access economic incentives. These incentives, detailed in subsequent sections of this paper, are allocated on the basis of the REC production plants. They include:

- a *valorization fee* determined by ARERA (the Italian Regulatory Authority for Energy, Networks, and Environment) that reimburses specific tariff components, such as transport and distribution costs, for self-consumed electricity; and
- a *premium tariff* applied to energy that is both self-produced and consumed within the REC.

Additionally, REC producers can monetize surplus energy fed into the grid by selling it.

The GSE, which is wholly owned by the Italian government via the Treasury, manages and monitors the incentive system. Its primary role is to oversee Italy's incentive framework, promoting the adoption of green technologies while ensuring compliance and effective implementation.

3.2. Economic model

To assess the profitability of a PV plant, the discounted cash flow (DCF) model was employed. This method assesses an investment by analyzing the difference between revenues (discounted cash inflow [DCI]) and costs (discounted cash outflow [DCO]), with both variables discounted over the project's useful lifetime [25,57].

$$NPV = DCI - DCO \quad (1)$$

In the present analysis, incoming cash flows consisted of several revenue items: (i) savings on the energy bill, resulting from the percentage of self-consumption and the purchase price of energy; (ii) the sale of excess energy not consumed to the grid; (iii) subsidies for self-consumption, resulting from the produced and self-consumed energy and the unit subsidy; (iv) benefits from the return of tariff components according to the self-consumed energy; and (v) tax deductions applied to the initial investment.

$$DCI = \sum_{t=1}^N \frac{\omega_{self,c} \times E_{Out,t} \times p_t^c + (1 - \omega_{self,c}) \times E_{Out,t} \times p_t^s + \omega_{self,c} \times E_{Out,t} \times S_{REC}^u + C_{val,t} \times \omega_{self,c} \times E_{Out,t}}{(1+r)^t} + \sum_{t=1}^{N_{TaxD}} \frac{\left(\frac{C_{inv}}{N_{TaxD}}\right) \times TaxD_u}{(1+r)^t} \quad (2)$$

$$E_{Out,t} = t_r \times k_f \times \eta_{lm} \times \eta_{bos} \times A_{cell} \times S \quad (3)$$

$$E_{Out,t+1} = E_{Out,t} \times (1 - dE_f) \quad (4)$$

$$p_{t+1}^c = p_t^c \times (1 + inf_{el}) \quad (5)$$

Cash outflows encompassed all expenditures associated with the PV system throughout its useful lifetime: (i) the initial investment cost; (ii) maintenance costs; (iii) assurance costs; (iv) taxes on energy sales; (v) contributions to the GSE; (vi) inverter replacement after 10 years of operation and (vii) administrative and connection costs.

$$DCO = \sum_{t=0}^{N_{debt}-1} \frac{\frac{C_{inv}}{N_{debt}} + (C_{inv} - C_{lcs,t}) \times r_d}{(1+r)^t} + \sum_{t=1}^N \frac{P_{Cm} \times C_{inv} \times (1 + inf) + P_{Cass} \times C_{inv} \times (1 + inf) + Sp_{el,t} \times P_{Ctax}}{(1+r)^t} + \sum_{t=1}^N \frac{T_{GSE} + T_{GSE,rid}}{(1+r)^t} + \frac{P_{Ci} \times C_{inv}}{(1+r)^{10}} + C_{ae} \quad (6)$$

$$C_{inv} = C_{inv,unit} \times (1 + Vat) \times S \quad (7)$$

3.3. Identification of scenarios

The REC analysis was conducted under two distinct market scenarios and one policy scenario. The two market scenarios were based on the purchase price of electricity, due to its significant variation and critical influence on the economic value of self-consumption. These scenarios were based on the observed trends in electricity costs during 2023 and the first quarter of 2024 for residential utilities:

Table 1
Input data [25,32,60–65].

Acronym	Value	Acronym	Value
A _{cell}	7 m ² /kWp	P _{Cl}	15 %
C _{ae}	1000 €	P _{Cm}	2 %
C _{inv,unit}	1850 €/kW	P _{Ctax}	27.5 %
C _{val,t}	0.01057 €/kWh	p ^{ex}	0.20 €/kWh
dE _f	0.80 %	p ^s	0.10 €/kWh
inf	3 %	r	5 %
inf _{el}	3 %	r _d	3 %
k _f	1.13	S	20 kW
N	20 years	S _{REC} ^u	0.104 €/kWh
N _{debt}	10 years	t _r	1450 kWh/m ²
N _{TaxD}	10 years	TaxD _u	50 %
η _{bos}	85 %	Vat	10 %
η _{lm}	20 %	wself _c	0%–100 %
p ^c	0.25–0.40 €/kWh	T _{GSE}	14 €
P _{Cass}	1 %	T _{GSE,rid}	15 €

- Low-market (LM) scenario: purchase cost of energy (p^c) set at 0.25 €/kWh.
- High-market (HM) scenario: purchase cost of energy (p^c) set at 0.40 €/kWh.

In both scenarios, the energy sale price to the grid (p^s) was fixed at

0.10 €/kWh, as determined by monthly average price trends in the “center-south” market area during the last half of 2023 and the first quarter of 2024, based on data collected from the GME website.

The policy scenario incorporated the 2023 Decree promoted by the Italian Ministry of the Environment and Energy Security (MASE). The incentive tariff on renewable energy produced and shared within the REC was assumed equal to 0.104 €/kWh (S_{REC}^u), as derived from the average of the minimum value of 0.084 €/kWh and the maximum value of 0.124 €/kWh. The reference values of 0.080 and 0.120 €/kWh were adjusted by the bonus of 0.004 €/kWh applicable to installations located in central Italy. Finally, a 50 % tax deduction was applied over a 10-year period, consistent with values frequently proposed in the literature. This

allowed for comparison with alternative contexts featuring different incentives or tax policies [42,58,59].

The case analyses were conducted across 11 self-consumption configurations (ω_{self,c}), varying between 0 % and 100 % with a delta of 10 %. These rates were calculated on an annual basis, in alignment with the literature [25].

Sensitivity, scenario and risk analyses were performed to enhance the robustness of the results and generate deeper insights into the impact of critical variables on the NPV.

3.4. Input data

The project focused on the installation of a 20 kW PV system to

establish an REC in a residential context within central Italy. This system would be designed to meet the energy requirements of four consumers, acting as RSCs, in a location characterized by an average annual insolation of approximately 1450 kWh/m². The system would be installed in a cluster of cottages, with four cottages collectively contributing to the project via a 5 kW PV system. The system output was assumed to begin at 38,996 kWh in the first year, decreasing annually by 0.80 % due to the natural degradation of the PV modules and other system components. By the 20th year, production was estimated to drop to 33,477 kWh. The system was expected to operate effectively for 20 years. The project would involve total financing through a third party, with a 10-year debt amortization period. In addition, the economic analysis considered a general and energy-specific inflation rate of 3 % per year.

The unit investment cost was estimated at 1850 €/kW, alongside ancillary costs encompassing insurance (1 % of the total cost), inverter replacement after 10 years (15 % of the investment cost) and maintenance costs (2 % of the total cost). Taxes were assumed to be 27.5 %. The project also incorporated an administrative and grid connection cost of 1000 €, incurred at year zero. On the revenue side, components such as bill savings, energy sales, tax deductions and subsidies, as outlined in Section 3.2, were assumed to be supplemented by an additional incentive of 0.01057 €/kWh for self-consumed energy, which would contribute to improving the profitability of the investment. Table 1 proposes the economic and technical parameters of the project.

3.5. New economic models for the distribution of profits from a PV system

In accordance with the literature, two profit distribution models (PDM) were initially considered for the distribution of benefits [25,32,65]:

- “Revenues split equally” (PDM1), ensuring an equal distribution of NPV among all RSCs, irrespective of their individual energy consumption; and
- “Revenues shared entirely according to energy consumption profile” (PDM2), with NPV allocated on the basis of each RSC's energy consumption profile, giving greater weight to more virtuous and synchronous behaviors between energy production and demand.

An intermediate approach discussed in the literature is based on a partial energy consumption profile, with the exchange price set as the average of the energy purchase and selling prices. In this paper, two new profit distribution models are proposed, tailored to specific needs:

- “Revenue allocation based on partial energy consumption and virtuous behavior” (PDM3), considering the partial energy consumption of individual RSCs, awarding a premium to those who propose lower trading prices. This model simulates an auction scenario in which the lowest bid (declared during REC establishment) wins.
- “Revenue allocation based on partial energy consumption with a bonus for lower incomes” (PDM4), prioritizing equity by awarding bonuses to RSCs with lower incomes. While partial energy consumption is still considered, the focus shifts to supporting individuals who are more vulnerable due to rising energy costs and at risk of energy poverty. Income brackets are declared during the establishment of the REC and are assumed to remain unchanged through the plant's lifetime, though annual reviews are required.

This paper provides a detailed analysis of the proposed benefits-sharing models within the framework of the two new PDMs. While eqs. (8) through (23) remain consistent across both models, the weight assigned to each RSC in eq. (17) differs, leading to distinct benefits-maximization approaches: PDM3 incentivizes virtuous behavior

during the exchange price proposal stage, whereas PDM4 prioritizes equity by considering the income brackets of individual prosumers. Both models follow the same initial step: calculating the average self-consumption value among all RSCs by summing their individual values and dividing by the total number of RSCs.

$$\omega_{\text{self},cj} = \frac{\sum_{j=1}^{N_{\text{RSC}}} \omega_{\text{self},cj}}{N_{\text{RSC}}} \quad (8)$$

The energy produced by each RSC is calculated by dividing the total energy produced by the PV system by the number of RSCs.

$$E_{pj} = \frac{E_{\text{out}}}{N_{\text{RSC}}} \quad (9)$$

Subsequently, the energy partially self-consumed by each RSC is calculated as the product of energy produced and the minimum value between the average self-consumption value and the specific self-consumption value of each RSC.

$$E_{\text{PSC}j} = E_{pj} \times \min[\omega_{\text{self},cj}; \omega_{\text{self},cj}] \quad (10)$$

The partially non-self-consumed energy by each RSC is calculated by multiplying the energy produced by the minimum value between the average self-consumption percentage and the non-self-consumption percentage.

$$E_{\text{PNSC}j} = E_{pj} \times \min[\omega_{\text{self},cj}; 1 - \omega_{\text{self},cj}] \quad (11)$$

Economic savings from partial self-consumed energy are then calculated as the product of the self-consumed energy and the purchase price of energy.

$$S_{\text{SC}j} = E_{\text{PSC}j} \times p^c \quad (12)$$

The calculation of economic incentives for self-consumption consists of two parts: (i) an individual incentive linked to self-consumed energy and (ii) a collective share distributed equally among all RSCs based on traded energy.

$$S_{\text{S}j} = S_u \times E_{\text{PSC}j} + \frac{1}{N_{\text{RSC}}} \left[S_u \times \sum_{j=1}^{N_{\text{RSC}}} (E_{pj} - E_{\text{PSC}j} - E_{\text{PNSC}j}) \right] \quad (13)$$

This formula represents the revenue from the sale of the partial non-self-consumed energy, calculated as the product of the partial non-self-consumed energy and the energy sale price.

$$S_{\text{Net}j} = E_{\text{PNSC}j} \times p^s \quad (14)$$

During the establishment of the charter underlying the REC, each member proposes an exchange price. The exchange price implemented within the community is then determined as the average of the prices proposed by the various RSCs.

$$p^{\text{ex}} = \frac{\sum_{j=1}^{N_{\text{RSC}}} p_p}{N_{\text{RSC}}} \quad (15)$$

The benefits from energy exchange are divided into two components, based on two distinct distributions. Fifty percent of the benefits are distributed based on the energy exchanged at the exchange price.

$$S_{\text{EX}j} = \frac{1}{2} [(E_{pj} - E_{\text{PSC}j} - E_{\text{PNSC}j}) \times p^{\text{ex}}] \quad (16)$$

The remaining 50 % is distributed according to the virtuous behavior of RSCs. A weight (p_{vj}) is calculated to reflect the virtuous behavior of an RSC, defined as the ratio of the exchange price to the price proposed by each RSC.

$$p_{vj} = \frac{p^{\text{ex}}}{p_p} \text{ con } j = 1 \dots N_{\text{RSC}} \quad (17)$$

Subsequently, this weight is transformed into a percentage for each RSC as the ratio of the weight of the j -th RSC to the sum of all RSC weights.

$$\omega_{vj} = \frac{p_{vj}}{\sum_{j=1}^{N_{RSC}} p_{vj}} \text{ con } j = 1 \dots N_{RSC} \quad (18)$$

To calculate the benefits distribution, the savings from the exchange are multiplied by the benefits percentage.

$$S_{EXvj} = \left(\sum_{j=1}^{N_{RSC}} S_{EXj} \right) \times \omega_{vj} \quad (19)$$

The total benefits for each RSC represent the sum of the savings from self-consumption, incentives for self-consumption, revenues from the sale of partial energy not self-consumed, savings from exchange and additional savings based on virtuous behavior.

$$OB_j = S_{SCj} + S_{Sj} + S_{Netj} + S_{EXj} + S_{EXvj} \quad (20)$$

The total overall benefits is calculated as the total benefits for all RSCs.

$$OB_{TOT} = \sum_{j=1}^{N_{RSC}} OB_j \quad (21)$$

To calculate the benefits percentage for each RSC, the ratio of each RSC's total benefits to the overall total benefits is computed.

$$PD_{OVj} = \frac{OB_j}{OB_{TOT}} \quad (22)$$

Finally, the NPV associated with each RSC is determined by multiplying the total NPV by the benefits percentage of each RSC.

$$NPV_{RSCj} = NPV \times PD_{OVj} \text{ With } j = 1 \dots N_{RSC} \quad (23)$$

Concerning PDM3, as an initial example, the model was analyzed for four RSCs. Within a broader research project, a survey was conducted in May 2024 evaluating 402 Italian citizens with an average age of 37 years. To enhance the realism of the simulation, the responses considered were those from individuals interested in participating in an RSC after receiving an explanation of this concept. This narrowed the analysis to 276 respondents. To identify four potential RSC profiles, the profiles from the survey were ranked according to the exchange price proposed by each RSC. Subsequently, they were sorted in ascending order and divided into four equally populated groups, with an average value calculated for each group:

- RSC1: 0.081 €/kWh.
- RSC2: 0.168 €/kWh.
- RSC3: 0.239 €/kWh.
- RSC4: 0.309 €/kWh.

In a REC, the exchange price (p^{ex}) was set to €0.20/kWh, based on the average input from 276 participants. For PDM4, while the exchange price remained fixed at €0.20/kWh, the primary innovation lay in the subjective assignment of weights. This adjustment was necessitated by the use of income data provided in ranges rather than precise values. To promote equity, the model assigned the highest weight to prosumers in the lowest income bracket, effectively incentivizing and rewarding participation for lower-income groups. The income brackets used for this analysis were: €0–20,000, €20,000–40,000, €40,000–60,000, and €60,000 and above.

A sensitivity analysis was conducted for both models by varying the self-consumption percentage of one RSC from 10 % to 90 %, while keeping the other three fixed. This approach aimed at uncovering the effects of self-consumption on the distribution of benefits within the community. This analysis was performed using two contrasting RSC

profiles: one with a high percentage of self-consumption and another with a low percentage.

Finally, a second application example was presented, incorporating an external consumer (a non-producer) into an existing REC. While the four RSCs were comprised of prosumers, in this example, the external consumer did not participate in the production of energy but merely purchased it from the community, taking equal shares from the four RSCs. In doing so, the external consumer benefited from a lower energy purchasing price than that of the traditional power grid, while the RSCs gained from a higher energy selling price than they could achieve from sale to the grid. For the external consumer, the benefit was calculated as:

$$S_{EX} = E_r \times p^c - E_r \times p^s \quad (24)$$

reflecting the economic savings achieved through participation in the REC.

4. Analysis and results: Photovoltaic system

The results presented below refer to the PV system within the baseline scenario (Section 4.1) and the alternative contexts (Section 4.2).

4.1. Results for the baseline scenario

The primary objective was to evaluate the profitability of PV plants. Using the economic model described in eqs. (1) to (7) and the input data provided in Table 2 and Fig. S3, the variation in NPV was calculated across the different case studies, considering the percentage of self-consumption as a critical variable. Twenty-two case studies were analyzed, representing combinations of the possible self-consumption shares and market scenarios.

Profitability was verified in most scenarios, except when energy was entirely sold to the grid, corresponding to 0% self-consumption. A break-even analysis was conducted, yielding values of 2.4 % for the high-market (HM) scenario and 3.7 % for the low-market (LM) scenario. These values are slightly higher than those reported in previous analyses, primarily due to the lower energy purchase prices, which reduced the revenues generated from savings in the energy bill [25]. Since the cost of purchasing energy from the grid exceeded the revenue obtained from its sale, an increase in self-consumption resulted in significant expense savings that outweighed the reduction in profit from grid sales. Consequently, profitability was strongly correlated with the percentage of self-consumption. Previous analyses have suggested that self-consumption rates in the residential sector typically range from 30 to 70 % [25,32,62,66,67] (Fig. 3).

NPV ranged from 2041 €/kW in the LM scenario with 30 % self-consumption to 8195 €/kW in the HM scenario with 70 % self-consumption. A clear trend emerged, showing that, for every 10 % increase in self-consumption, NPV rose by 1212 €/kW in the HM scenario and 777 €/kW in the LM scenario. In addition, the disparity in NPV between the LM and HM scenarios grew as the percentage of self-consumption increased. The adoption of energy consumption practices

Table 2
NPV (€), Baseline scenario.

% Self-consumption	HM scenario	LM scenario
0%	–5767	–5767
10 %	18,471	9765
20 %	42,710	25,298
30 %	66,949	40,831
40 %	91,189	56,364
50 %	115,428	71,897
60 %	139,667	87,430
70 %	163,906	102,963
80 %	188,145	118,496
90 %	212,384	134,029
100%	236,623	149,562

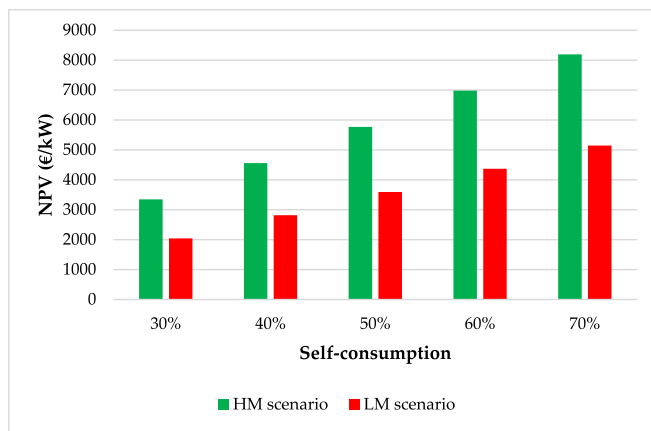


Fig. 3. NPV (€/kW), Baseline scenario.

Table 3

Literature review, NPV.

Indicator	Value	References
NPV	1293–8537 k€	[70]
NPV	233–769 k€	[71]
NPV	1300–2119 €/kW	[53]
NPV	3.21–21.35 M€	[72]
NPV	12–38,702 €	[73]
NPV	2953–9584 €/kW	[25]
NPV	90–140 k€	[74]
NPV	74–248 k€	[75]
NPV	(–2317)–(–399) k€	[31]
NPV	(–254)–1211 k€	[76]
NPV	2136–8084 €/kW	[32]
NPV	40–52 k€	[77]
NPV	50–1174 k€	[78]
NPV	1035–3178 €/kW	[79]
NPV	33–52	[80]
NPV	700–1400 k€	[81]

that aligned with production phases (i.e., virtuous behavior) enhanced self-consumption, thereby improving profitability. This effect became especially pronounced in market contexts with higher energy purchase prices, as the installation of PV systems significantly amplified prosumer benefits. However, the observed dynamic must be carefully considered by policymakers to prevent or mitigate potential energy poverty issues.

The present results are less favorable than the results reported in the literature. For example, in a previous analysis [25], NPV values for a 30 % self-consumption rate ranged between 2953 and 3244 €/kW in the LM scenario and 4354–4645 €/kW in the HM scenario. The primary factor for this disparity was not the selling price or subsidy, but the purchase price, which was set to 350 €/MWh and 500 €/MWh in the respective LM and HM scenarios. A similar trend was observed in another study [32], in which NPV ranged between 1961 and 2136 €/kW in the LM scenario and 4713–4888 €/kW in the HM scenario. The greater NPV difference in the HM scenario stemmed primarily from variations in the energy selling price.

A comparison with the literature is crucial, and Table 3 presents the NPV outcomes across diverse market and policy contexts. The key insight from this comparison highlights the importance of conducting analyses in alternative contexts to validate or challenge the findings of the baseline scenario. However, while NPV is a key metric, other economic indicators may provide useful insights. For instance, economic savings associated with the installation of RECs has been shown to range from 32 % [68] to between 1.6 and 19.5 % [46]. In a separate work, the levelized cost of electricity (LCOE) was found equal to 114–263 €/MWh [69]. In the present study, the LCOE was lower and equal to 105 €/MWh, consistent with previous analyses [25] (though slightly higher than the

94 €/MWh reported in one study [32]).

Tax deductions also play a pivotal role in shaping economic outcomes, contingent on the assumption that prosumers possess sufficient tax capacity to fully utilize these benefits. This becomes more feasible when deductions are spread over 10 years. However, such policies may disadvantage low-income individuals, who may lack the financial capacity to fully benefit from these deductions. A comparison with the literature reveals that increasing the tax deduction rate from 36 % (baseline) to 50 % could raise the NPV by approximately €216/kW [32], while further increasing the rate to 90 % could boost the NPV by an additional €908/kW. This highlights the significance of tax deduction rates on economic outcomes and the need for equitable policy adjustments [65].

4.2. Results for the alternative scenarios

To strengthen the findings for the baseline case, alternative scenarios were analyzed, focusing on critical variables identified in the literature [25,82,83]. Specifically, the analysis considered variations of 50 €/MWh in the energy purchase price, 50 €/MWh in the energy sale price and 200 €/kW in the unit investment cost. These scenarios were examined across self-consumption percentages in the range of 30–70 %.

4.2.1. Sensitivity analysis

A sensitivity analysis was conducted to evaluate the impact of varying a single variable at a time. In total, 60 cases were studied (Table 4 and Fig. S4a–b).

The results confirmed that profitability was consistently achieved. However, it is notable that NPV reached its minimum in both the LM and HM scenarios when the selling price decreased by 0.05 €/kWh. At 30 % self-consumption, the corresponding NPV values were 1305 €/kW in the LM scenario and 2611 €/kW in the HM scenario. In contrast, NPV reached its maximum in both scenarios when the energy purchase price increased by 0.05 €/kWh at 70 % self-consumption (6164 €/kW and 9211 €/kW, respectively).

The disparity between the HM and LM scenarios widened as self-consumption increased. This could be attributed to the growing proportion of energy produced and consumed directly by prosumers, which enhanced profitability. From a quantitative perspective, a 0.05 €/kWh change in the energy purchase price resulted in an NPV variation of 435–1015 €/kW. Similarly, a 0.05 €/kW change in the selling price corresponded to an NPV variation of 315–737 €/kW, while a 200 €/kW change in the unit investment cost led to an NPV change of 251 €/kW.

4.2.2. Scenario analysis

A scenario analysis was conducted by simultaneously varying all

Table 4

NPV (€/kW), sensitivity analysis.

	Pessimistic	Optimistic	Pessimistic	Optimistic
	Investment cost scenario HM		Investment cost scenario LM	
30 %	3096	3599	1790	2293
40 %	4308	4811	2567	3069
50 %	5520	6022	3343	3846
60 %	6732	7234	4120	4623
70 %	7944	8446	4897	5399
	Purchase price scenario HM		Purchase price scenario LM	
30 %	2912	3783	1606	2477
40 %	3979	5140	2238	3399
50 %	5046	6497	2869	4320
60 %	6113	7854	3501	5242
70 %	7179	9211	4132	6164
	Selling price scenario HM		Selling price scenario LM	
30 %	2611	4084	1305	2777
40 %	3928	5190	2187	3449
50 %	5245	6297	3069	4120
60 %	6562	7404	3951	4792
70 %	7880	8511	4833	5463

Table 5
NPV (€/kW), scenario analysis.

	Pessimistic		Optimistic	
	HM scenario	LM scenario	HM scenario	LM scenario
30 %	1925	619	4770	3464
40 %	3097	1355	6022	4281
50 %	4269	2092	7274	5098
60 %	5441	2829	8526	5914
70 %	6613	3566	9778	6731

three critical variables, exploring both optimistic and pessimistic cases. In total, 20 cases were analyzed (Table 5 and Fig. S5).

The results confirmed the profitability of the analyzed case studies. Specifically, in the optimistic scenario, each 10 % increase in self-consumption was associated with an NPV increase of 1252 €/kW in the HM scenario and 816 €/kW in the LM scenario. In the pessimistic scenario, these changes were slightly lower, at 1172 €/kW in the HM scenario and 736 €/kW in the LM scenario. These results confirmed that, even under the most unfavorable conditions, the PV system delivered positive benefits, though the magnitude of returns was sensitive to fluctuations in key variables.

4.2.3. Risk analysis

Finally, a risk analysis was conducted using the Monte Carlo method, with 1000 iterations performed to estimate NPV. The simulations used the baseline mean values for the three key variables (i.e., energy selling price, energy purchase price, unit investment cost), and derived standard deviations from the variance used in the alternative scenarios. The results showed a 100 % probability of achieving a positive NPV across all case studies (Fig. S6a–b), with one exception: in the LM scenario with 30 % self-consumption, the probability dropped to 98.9 % (Fig. 4).

The risk analysis for the extreme self-consumption rates of 30 % and 70 % revealed the following NPV distributions:

- 30 % self-consumption and LM scenario: 39 % of values were concentrated in the 20–40 k€ range and 36 % in the 40–60 k€ range.
- 30 % self-consumption and HM scenario: 45 % of values were concentrated in the 60–80 k€ range and 26 % in the 40–60 k€ range.

- 70 % self-consumption and LM scenario: 32 % of values were concentrated in the 80–100 k€ range and 31 % in the 100–120 k€ range.
- 70 % self-consumption and HM scenario: 34 % of the values were concentrated in the 160–180 k€ range and 28 % in the 140–160 k€ range.

These results reaffirmed the investment's profitability, demonstrating an almost negligible economic risk.

5. Analysis and results: Renewable energy community

The revenue distribution patterns within the REC were analyzed by comparing the baseline scenario (Section 5.1) with the alternative scenarios (Section 5.2). Additionally, the evaluation considered the inclusion of a consumer within the REC framework (Section 5.3).

5.1. Distribution of benefits within the REC: Baseline scenario

The second analysis aimed to evaluate the distribution of benefits among the four RSCs, according to eqs. (8)–(24). It was assumed that the RSCs would share the benefits derived from the PV installation and their varying consumption habits would contribute to an average self-consumption rate of 50 %. The following consumption profiles were assumed: 80 %, 65 %, 35 % and 20 % for RSC1, RSC2, RSC3 and RSC4,

Table 6
Revenues by energy consumption profile (PDM2).

	RSC1	RSC2	RSC3	RSC4	Total
Self-consumption percentage	80 %	65 %	35 %	20 %	
Energy produced (kWh)	9749	9749	9749	9749	38,996
Self-consumed energy (kWh)	7799	6337	3412	1950	19,498
Energy not self-consumed (kWh)	1950	3412	6337	7799	19,498
Self-consumption incentives (€)	893	726	390	223	2233
Savings in utility bills (€)	3120	2535	1365	780	7799
Gain from sale (€)	195	341	634	780	1950
Total benefits (€)	4208	3602	2389	1783	11,983
Percentage distribution of benefits	35.1 %	30.1 %	19.9 %	14.9 %	100 %

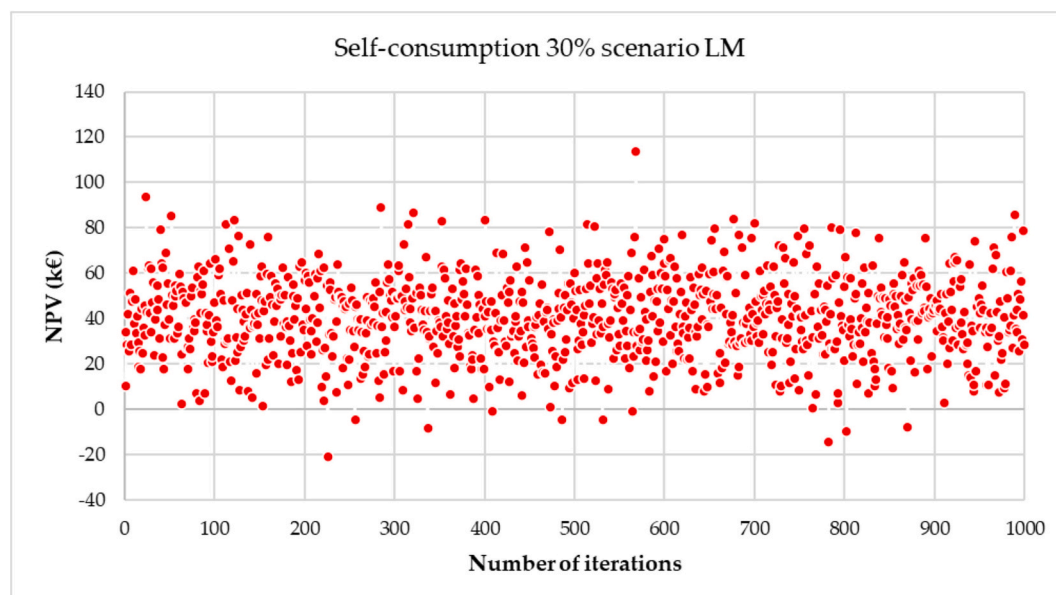


Fig. 4. NPV (k€), risk analysis.

respectively. The PDM for each RSC was calculated by multiplying their revenue-sharing percentage by the obtained NPV. This section considers the HM scenario, with four distinct PDM models proposed, as outlined in Section 3.

The first PDM, “revenues split equally,” divided the total NPV by the number of RSCs. With four RSCs, each was allocated 25 % of the total revenues, irrespective of their specific consumption profile.

The second PDM, “revenues shared entirely according to energy consumption profile,” considered the total energy produced by the PV system (38,996 kWh) and the varying consumption profiles (Table 6). Each RSC benefited through three revenue streams. For example, RSC2, with 65 % self-consumption, self-consumed 6337 kWh, resulting in an estimated economic savings of 2535 € and an economic benefit of 893 € from incentives for self-consumed energy. The remaining 35 % of the energy not self-consumed was sold to the grid, generating additional revenues of 341 €. The sum of these three values (3602 €) was divided by the total REC benefits (11,983 €), yielding a benefits distribution of 30.1 % for RSC2. By contrast, RSC3, characterized by a self-consumption rate of 35 %, achieved a benefits distribution is 19.9 %. In scenarios where lower purchase prices were assumed, the economic savings from electricity bill reductions were diminished, leading to a smaller proportional share of benefits for RSCs with lower consumption profiles.

The third PDM, which allocated revenues based on a combination of partial energy consumption and virtuous behavior, introduced a novel apportionment approach not previously explored in the literature. This model served as an intermediate solution between the two earlier models while simultaneously incentivizing RSCs to adopt energy-efficient and synchronized consumption behaviors (Table 7).

Unlike the previous PDMs, this approach assumed that each RSC was entitled to a maximum self-consumption share set at 50 % (the average self-consumption rate among the four RSCs). For example, RSC2, with a self-consumption rate of 65 %, was limited to accessing 50 % of its self-produced energy. This adjustment resulted in economic savings of 1950 €. The remaining 35 % of the self-produced energy was sold to the grid, generating additional revenue of 341 €. The remaining 15 % was exchanged with another RSC at an exchange price determined as the average of the values proposed by the four RSCs during the drafting of the REC statute.

The exchange price was set at 0.20 €/kWh. Total benefits were divided into two parts. The first component evaluated half of the energy exchanged by each RSC at the fixed exchange price, as proposed in the literature [25]. The second component applied a weighted value calculated based on the difference between the price proposed by each RSC and the average exchange price. To illustrate the calculation of the

percentage benefits distribution, consider RSC2, whose revenues were derived as follows:

- (i) incentives from self-consumption were obtained by multiplying 4874 kWh by the unit incentive of 0.11457 €/kWh ($0.104 + 0.01057$). An additional 8774 kWh was also considered, with the same unit incentive (0.11457 €/kWh), divided by four (the number of RSCs). The first component amounted to 558 €, while the second was 251 €, resulting in a total incentive value of 809 €.
- (ii) Bill savings were derived from 4874 kWh at a rate of 0.40 €/kWh, amounting to 1950 €.
- (iii) Revenue from energy sales was obtained by selling 3412 kWh at 0.10 €/kWh, resulting in 341 €.
- (iv) Energy exchange revenue was calculated from 50 % of 1462 kWh at the exchange price of 0.20 €/kWh, yielding 146 €.
- (v) The incentive exchange was determined from 50 % of 8774 kWh at the exchange price of 0.20 €/kWh, multiplied by a premium weight of 0.23, equaling 202 €.

The sum of these five contributions was 3449 €, representing 28.6 % of the total benefits, amounting to 12,047 €.

This model may lack robustness against collusion, as participants could strategically offer lower prices to gain a disproportionate weight in the distribution of rewards.

Given that RECs are also designed to combat energy poverty, a correction to the PDM3 model was proposed, resulting in the development of PDM4. This revised model allocated benefits based on partial energy consumption profiles, with an added bonus for lower-income participants. Instead of relying on the proposed exchange price, PDM4 used income brackets, assigning higher weights to participants in lower income ranges. These weights gradually decreased as income bracket rose. In this context, the assignment of weights was subjective and structured as follows:

- RSC1, with an income bracket of 60,000+ €, had a weight of 0.10;
- RSC2, with an income range of 40,000–60,000 €, had a weight of 0.20;
- RSC3, with an income range of 20,000–40,000 €, had a weight of 0.30; and
- RSC4, with an income range of 0–20,000 €, had a weight of 0.40.

The incorporation of this mechanism to safeguard economically disadvantaged participants resulted in the new benefits distribution illustrated in Fig. 5. Under this new model, the range of benefits distribution percentages across the four RSCs narrowed significantly compared to the PDM3 model based on virtuous behavior (19.80–28.41 % vs. 17.84–30.45 %). Of note, RSC1 in PDM3 corresponded to RSC1 in PDM4 only in terms of the energy profile, as the proposed exchange price and income levels were not directly correlated in PDM4.

However, a critical observation emerged from the results: RSC2, despite having a lower self-consumption percentage and potentially qualifying for fewer benefits than RSC1, received a larger share of benefits due to its lower income and the corresponding bonus. This demonstrates the significant impact of the income-based bonus on the overall benefits distribution. The result was also heavily influenced by the weight ranges used in PDM4, which spanned 10–40 % (compared to the 13–48 % range in PDM3). Finally, both PDM3 and PDM4 produced intermediate benefits distributions among the four RSCs when compared to the two extreme models, PDM1 and PDM2.

5.2. Distribution of benefits within the REC: Alternative scenarios

To enhance the robustness of the results, a sensitivity analysis was conducted. This analysis focused on the self-consumption percentage of

Table 7
Breakdown based on partial energy consumption and virtuous behavior (PDM3).

	RSC1	RSC2	RSC3	RSC4	Total
Self-consumption percentage	80 %	65 %	35 %	20 %	
Energy produced (kWh)	9749	9749	9749	9749	38,996
Self-consumed energy (kWh)	7799	6337	3412	1950	19,498
Energy not self-consumed (kWh)	1950	3412	6337	7799	19,498
Partial self-consumed energy 50 % (kWh)	4874	4874	3412	1950	15,111
Partial energy not self-consumed (kWh)	1950	3412	4874	4874	15,111
Energy exchange (kWh)	2925	1462	1462	2925	8774
Energy exchange weight (%)	0.48	0.23	0.16	0.13	100
Self-consumption incentives (€)	809	809	643	474	2736
Savings in the utility bill (€)	1950	1950	1365	780	6044
Gain from sale (€)	195	341	487	487	1511
Energy exchange (€)	292	146	146	292	877
Incentive exchange (€)	421	202	140	115	877
Total benefits (€)	3668	3449	2781	2149	12,047
Percentage distribution of benefits	30.5 %	28.6 %	23.1 %	17.8 %	100%

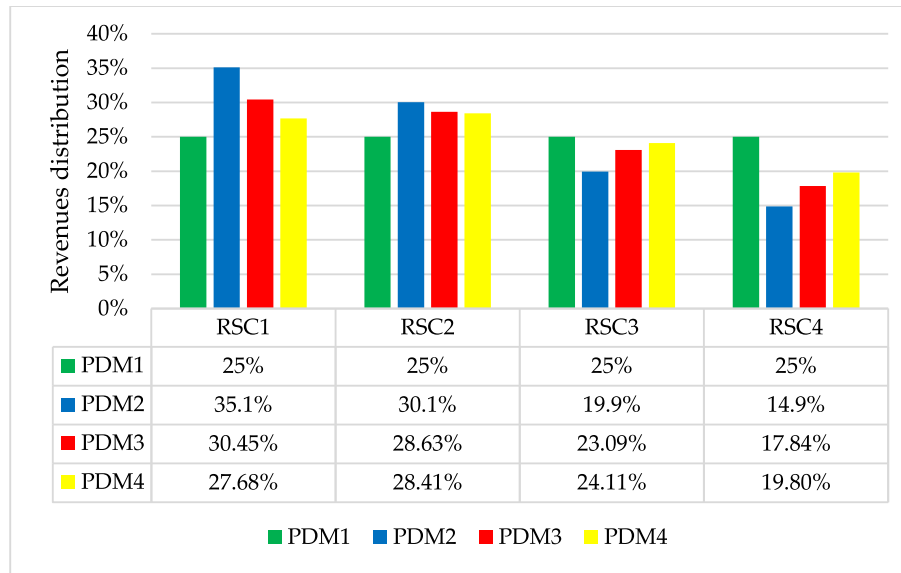


Fig. 5. Benefits distribution in different profit distribution models.

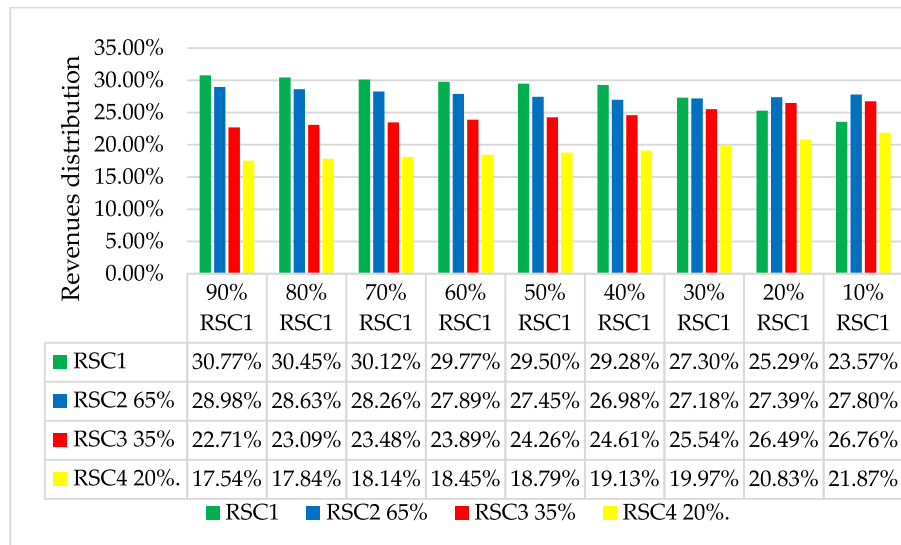


Fig. 6. PDM3, RSC1 sensitivity analysis.

RSC1 as the key variable to evaluate its impact on the distribution of benefits among the four RSCs under PDM3. The self-consumption percentage was varied within a range of 10–90 %, with a delta of 10 %, as the extreme cases of 0 % and 100 % self-consumption were considered highly unlikely according to the literature (Fig. 6).

As expected, given that the mechanism was based on a partial consumption profile (defined as the average of the various consumption profiles), a decrease in RSC1's self-consumption percentage negatively impacted RSCs with above-average consumption profiles. This occurred because a reduction in the partial consumption profile led to a proportional decrease in self-consumed energy for RSC2. Consequently, RSC2 experienced reduced bill savings, which were converted into benefits from energy exchange, though these benefits were valued at a lower price than the energy purchase price. Conversely, RSCs with below-average consumption profiles saw an increase in revenues. As the partial consumption profile decreased, the energy not partially self-consumed diminished, and this energy, when exchanged, was valued at a higher price than if sold to the grid. In general, the primary beneficiaries were RSC3 and RSC4, whose revenues grew steadily as RSC1's

profile decreased. RSC2, however, remained relatively stable, though a notable shift occurred when RSC1's self-consumption percentage decreased from 40 % to 30 %. During this transition, RSC1's reduced benefits resulted in additional benefits being allocated to RSC2. To ensure the robustness of these findings, the analysis was replicated for RSC4 (Fig. 7).

Examining the sensitivity analysis for RSC4 under the scenario in which weights were assigned according to the proposed price, it became evident that, even with increased self-consumption, RSC4 would never achieve the highest percentage of benefits due to its lower assigned weight (0.13). Across all of the nine case studies, RSC1 consistently secured the highest percentage of benefits, due to its combination of a significantly higher weight compared to the other RSCs and its very high self-consumption percentage. In contrast, RSC3 was most penalized as RSC4's self-consumption increased. This was because RSC3 had a lower self-consumption percentage relative to the partial profile accessed, particularly in later scenarios, and because its weight was lower than those of RSC1 and RSC2. Even though RSC3 had a slightly higher weight than RSC4 (0.16 vs. 0.13), this was insufficient to ensure higher gains

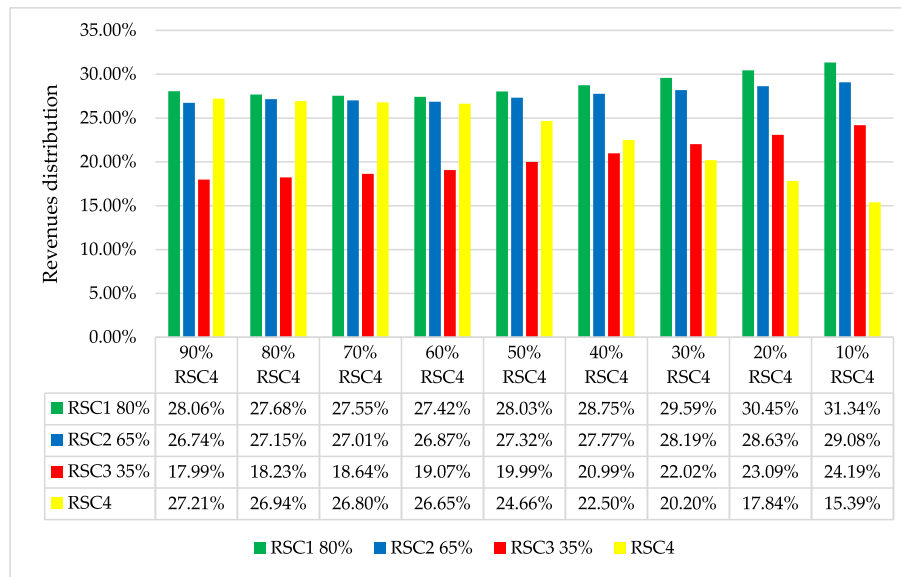


Fig. 7. PDM3, RSC4 sensitivity analysis.

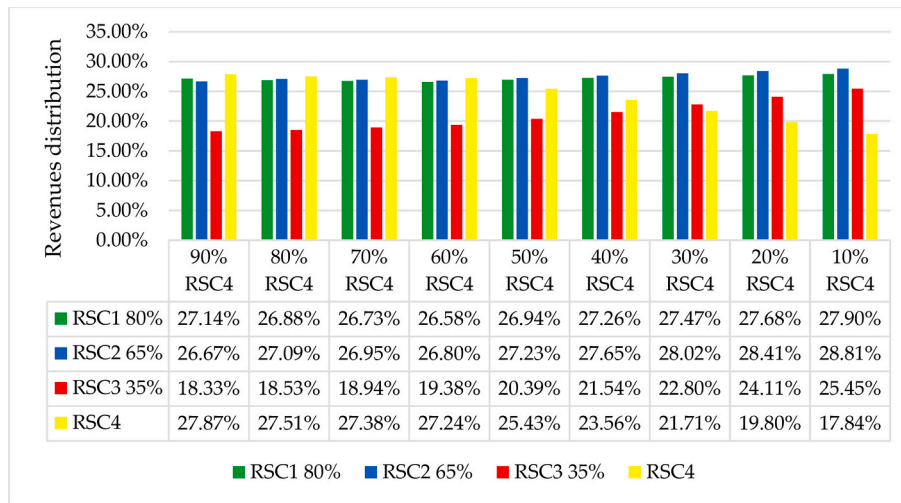


Fig. 8. Sensitivity analysis, PDM4-RSC4.

when RSC4 increased its self-consumption, precisely because of its significance in the allocation of gains.

Subsequently, PDM4 was examined. Again, a sensitivity analysis was conducted based on the self-consumption percentages of both RSC1 and RSC4. For RSC4 (Fig. 8), the lowest income and, consequently, the highest assigned weight positioned it to receive progressively more benefits as its self-consumption increased. This dynamic reduced the gap between RSC4 and the other RSCs. From 60 % self-consumption onward, RSC4 surpassed all other RSCs in benefits, even while self-consuming less energy than RSC1 and RSC2. Conversely, RSC3 did not benefit under this scenario. As RSC4's self-consumption increased, RSC3's benefits declined. This is because RSC3 was characterized by the lowest self-consumption percentage, which caused the partial profile to rise, favoring RSCs with higher self-consumption percentages, such as RSC1, RSC2 and RSC4.

On the other hand, examining the sensitivity analysis as a function of RSC1's self-consumption percentage (Fig. 9) revealed that, as RSC1's self-consumption decreased, partial energy consumption (calculated as the average of the self-consumption percentages of the four RSCs) also decreased. This shift primarily benefited RSC3 and RSC4, who had the

lowest self-consumption percentages but were still able to gain more than RSC2 due to the weight assigned to them based on income. Specifically, RSC3's share of benefits increased from 23.71 % to 28.85 %, while RSC4's share grew from 19.48 % to 25.92 %.

5.3. Distribution of benefits within the REC with the addition of a consumer

Finally, focusing on PDM3 and PDM4, we assumed that an additional consumer (C1) voluntarily joined the REC after its establishment. C1 participated solely as a consumer, purchasing energy from other members at an exchange price of 0.20 €/kWh, without contributing to production. C1's energy demand was assumed equal to 6000 kWh, representing 15.5 % of the total energy produced by the REC. This demand was met equally by the four RSCs, with each supplying 1500 kWh. The economic analysis revealed that each RSC increased its profits as subsidies rose, due to higher shares of self-consumed energy and exchange, alongside reduced shares of energy sold externally. The financial gains from subsidies and exchanges outweighed the losses associated with lower external sales. Therefore, the relative changes in

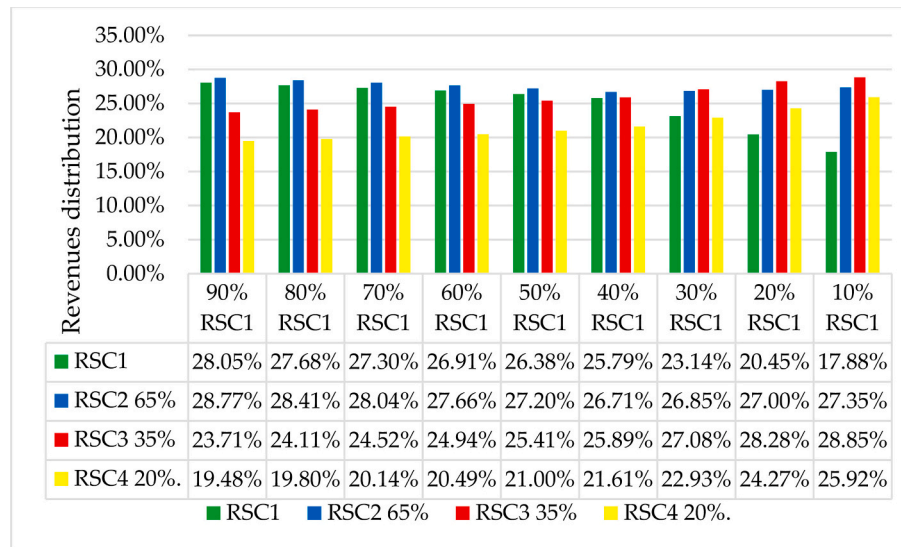


Fig. 9. Sensitivity analysis, PDM4-RSC1.

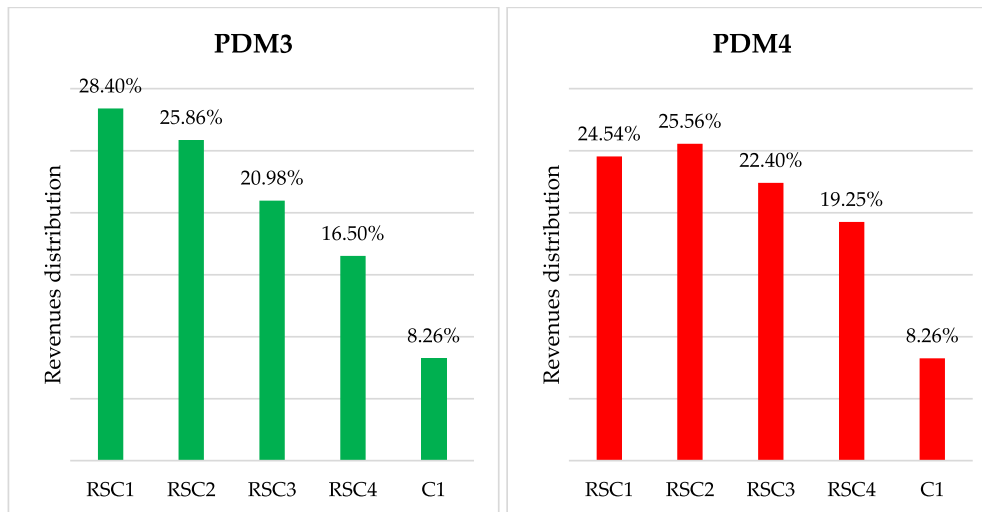


Fig. 10. Benefits distribution following the entry of a new consumer into the REC.

profit for each RSC were as follows:

- PDM3:
 - o RSC1: 3668€–4128€
 - o RSC2: 3449€–3758€
 - o RSC3: 2781€–3049€
 - o RSC4: 2199€–2398€
- PDM4:
 - o RSC1: 3335€–3556€
 - o RSC2: 3423€–3714€
 - o RSC3: 2904€–3255€
 - o RSC4: 2396€–2797€

C1 benefited from purchasing energy at a lower exchange price compared to standard market prices. This translated into savings of 1200 € for both PDM3 and PDM4, calculated as the difference between purchasing 6000 kWh at 0.40€/kWh and at 0.20€/kWh. The aggregate benefits of the RSCs and C1 are summarized and presented in Figs. 10 and S7.

The results show that C1 captured approximately 8 % of the total benefits in both PDM3 and PDM4, as their energy purchase from the REC

remained constant. In contrast, the benefits among the RSCs exhibited greater variability. In PDM3, the RSCs' share of benefits ranged from 16.50 to 28.40 %, while in PDM4, the range was narrower, varying between 19.25 and 25.56 %. These differences highlight the significant impact of the premium structure. In PDM3, RSC1 gained the largest share of benefits, reflecting its higher self-consumption rate and its willingness to offer a lower exchange price compared to other prosumers. In PDM4, although RSC1 continued to achieve the highest income due to its high percentage of self-consumption, its relative benefits were lower because the premium allocation favored RSC2, allowing the latter to claim the largest share of the benefits. Similarly, in PDM4, RSC3 and RSC4, despite being less efficient at synchronizing their consumption with energy production, benefited from a more equitable distribution of benefits due to the premium linked to minimum income levels.

6. Discussion and conclusion

Europe's ambitious pursuit of a climate-neutral continent represents an essential strategy for addressing the growing challenges posed by climate change. At the heart of this vision is the rapid expansion of renewable energy sources, and particularly PV systems, which are

fundamental to the continent's decarbonization goals. While large-scale renewable installations play a pivotal role in reducing carbon emissions, contributions from the residential sector are equally significant in driving this transition. Thus, a dual focus on both large-scale and residential energy generation forms the cornerstone of Europe's renewable energy strategy.

RECs represent a key mechanism for achieving this goal. Not only do they bolster national energy independence by reducing reliance on non-renewable sources, but they also foster deeper citizen engagement, encouraging the development of new social and economic structures aligned with the ecological transition. Crucially, however, this transition must be equitable. It should aim to include vulnerable populations—particularly those at risk of energy poverty—ensuring that the benefits of renewable energy are shared broadly across all socioeconomic groups. A just transition leaving no one behind is critical to the long-term success of Europe's green energy initiatives.

The present systematic literature review highlighted the growing importance of economic analyses and PV installations in driving the future of renewable energy. In this context, Europe's commitment to alternative energy strategies is not only significant but also imperative. Through policies that support the growth and development of RECs, Europe is positioning itself as a global leader in renewable energy innovation, demonstrating how collective action at the community level can contribute meaningfully to broader climate objectives.

6.1. Economic viability of PV systems in RECs

The primary finding of this study was the clear economic viability of PV systems within RECs, across varying market conditions. In the LM scenario, NPV values ranged from 2042 to 5148 €/kW, while in the HM scenario, they ranged from 3347 to 8195 €/kW. These calculations, based on self-consumption rates between 30 and 70 %, affirm the financial feasibility of REC-based PV installations. With a LCOE of 105 €/MWh, these systems demonstrate strong competitiveness against other energy sources.

The significance of these results cannot be overstated. As Europe accelerates its decarbonization efforts, residential PV systems within RECs present a viable pathway for reducing emissions while delivering substantial economic returns. In addition to cutting carbon emissions, these systems contribute to the creation of energy-efficient, economically sustainable communities. Furthermore, the sensitivity, scenario and risk analyses confirmed the profitability of these systems across a range of market conditions, highlighting their minimal economic risk. This robust financial performance makes REC-based PV systems an attractive investment opportunity for both governments and private individuals, reinforcing the role of residential solar power in achieving Europe's climate goals.

6.2. Distribution of benefits in RECs

The initial findings formed the basis for the second key finding: given the profitability of PV systems within RECs, the equitable and inclusive distribution of benefits among participants is critical, influencing REC adoption and long-term success. To address this challenge, the study introduced two innovative PDMs designed to allocate the economic benefits of PV systems most fairly. The first model employed an auction-based mechanism, incentivizing RSCs proposing lower energy exchange prices within the REC. This approach encouraged prosumers to offer competitive rates, benefiting the community as a whole. The second model emphasized economic inclusion by allocating a larger share of benefits to lower-income households, thereby directly tackling energy poverty and supporting prosumers with limited economic resources. Together, these models may provide novel pathways for advancing equity and inclusivity within RECs.

The adaptability of these PDMs is a key strength, enabling their application in diverse contexts. The analysis also encompassed scenarios

involving external consumers—individuals who did not produce energy but merely purchased it from the REC. This extended analysis demonstrated that the models may be implemented in real-world settings to maximize the financial benefits of energy sharing. By addressing the diversity in self-consumption rates and income levels among RSCs, these models offer a flexible framework for enhancing REC participation while promoting social equity and environmental sustainability.

6.3. Limitations and areas for improvement

Despite the promising results, several limitations of the present study should be acknowledged, particularly regarding the definition of weights used to reward virtuous behavior in the auction-based model. For instance, the online survey conducted as part of the research identified an average exchange price of 0.20 €/kWh, but this value may vary significantly depending on the respondent pool. Thus, the auction mechanism, while effective, may require contextual adaptation to reflect the behavior and preferences of different participant groups.

Similarly, the PDM rewarding lower-income households has inherent limitations due to its subjective approach to defining income brackets. A more robust mathematical framework could enhance the model by better accounting for income disparities, ensuring that the benefits are distributed in a manner that is both equitable and transparent. The present results suggest that a hybrid model, combining the auction mechanism with income-based rewards, could provide a more holistic solution. Such an approach would balance the need for economic efficiency with the imperative to promote social equity, ensuring that the economic benefits of RECs are shared broadly among participants. Additionally, while the study identified favorable policy and market conditions for REC development, including the critical role played by self-consumption, it also highlighted a key limitation: the need to explore social factors, beyond economic and environmental considerations. Indeed, understanding the social motivations that drive prosumers and consumers to engage with RECs will be essential for fostering participation and long-term success.

6.4. Broader implications for sustainable cities

The broader implications of this study suggest that the development of sustainable cities may be significantly enhanced through the integration of renewable energy infrastructure. However, to fully realize the potential of RECs, energy aggregation models are needed to foster social cohesion and community engagement. The success of RECs relies on a deep understanding of citizen behavior and motivations. Accordingly, social analyses and engagement strategies are integral to REC design, generating valuable insights into how best to encourage widespread participation.

The present study did not focus on a single case study, but instead assessed a wide range of scenarios through multiple analyses, ensuring comprehensive and robust findings. Regarding the two proposed PDMs, while they share a similar structure, they differ significantly in their weight assignment methodologies. The primary objective was to ensure the replicability of these models by clearly detailing all of the methodological steps employed. Although the literature on RECs and their economic analyses is extensive, the present work offers a novel contribution by introducing benefits distribution mechanisms that prioritize either lower energy exchange prices or greater inclusivity for lower-income participants, thereby addressing the critical issues of equity and inclusivity within RECs.

Moving forward, Europe's green energy transition will require not only an ideological commitment to sustainability, but also pragmatic approaches integrating sound economic models with clearly demonstrable benefits. As the present study has shown, RECs represent a vital tool for achieving the SDGs, and particularly SDG 7 (Affordable and Clean Energy) and SDG 11 (Sustainable Cities and Communities). By promoting decentralized, community-driven renewable energy systems,

RECs can become a cornerstone of the transition to sustainable and resilient urban environments.

6.5. Policy implications

The present study has provided valuable insights to guide the refinement of policies to better support RECs. It has suggested that, while the incentives offered by the Italian regulatory system are generally adequate to ensure the profitability of investments in RECs, the current framework appears overly generalized and somewhat open-ended. Specifically, the incentive structure applies uniformly across diverse geographical and social contexts, relying solely on the ability to share energy (i.e., consuming self-produced energy within the same REC). For instance, the system delegates decisions about supporting the most economically vulnerable users to the participants themselves, without mediating specific measures. Additionally, it does not prioritize the formation of RECs in areas in which they might achieve the greatest efficiency. Urban areas, and particularly congested cities with higher network development costs, are treated the same as less densely populated or rural areas, despite their differing challenges and opportunities.

Despite these limitations, the present study has offered a robust logical and quantitative framework that may serve as a foundation for testing innovative solutions. Such solutions could aim at maximizing cost-effectiveness not only for REC members but also for the broader community, thereby aligning economic benefits with societal needs.

6.6. Conclusion and future research directions

The present study has provided compelling evidence that PV systems within RECs are economically viable and hold significant potential to advance environmental sustainability and social equity. The innovative benefits distribution models introduced herein establish a framework for encouraging efficient energy behavior while addressing income disparities, ensuring that the benefits of renewable energy are equitably shared across communities.

To build on these findings, future research should focus on refining the parameters of the benefits distribution models, particularly by enhancing the mechanisms for rewarding virtuous behavior and addressing income inequality. Additionally, exploring the integration of diverse energy sources (e.g., wind, biomass) into RECs could further enhance the sustainability and resilience of urban energy systems. By expanding the scope of RECs to incorporate a broader array of renewable energy technologies, cities may progress more effectively towards their long-term climate goals while fostering greater social inclusion.

Finally, further empirical research into the social dimensions of RECs (e.g., citizen engagement, community participation) could yield valuable insights into strategies for encouraging their widespread adoption. By integrating technical, economic and social considerations, future studies may contribute to a more comprehensive and holistic understanding of how RECs can drive the creation of sustainable, resilient and inclusive cities.

CRediT authorship contribution statement

Paolo Basilio: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Alberto Biancardi:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Idiano D'Adamo:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation, Conceptualization. **Massimo Gastaldi:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Tan Yigitcanlar:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2025.125752>.

Data availability

Data will be made available on request.

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